MYSTERY METEORITE: PETROGRAPHIC INVESTIGATION OF AN UNCLASSIFIED CHONDRITE

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Abstract

Chondrites, the most primitive stony meteorites, offer invaluable insights into the early solar system's composition and evolution. Despite their significance, a considerable portion of specimens remain unclassified, with their origin and formative processes unknown. This study presents a petrographic investigation of an unclassified chondrite, aiming to elucidate its mineralogical and textural characteristics, as well as its potential implications for early solar system dynamics. The specimen, obtained from an online dealer and labeled as "NWA ?" (northwest Africa, source unknown) can be shown to be an undifferentiated stony meteorite based on external characteristics. The petrography of the specimen reveals a metamorphosed, Type V-VI, chondrite with relic chondrules and substantial evidence of terrestrial weathering. This type of meteorite derived from a parent body that experienced substantial heating during solar system accretion, but one that did not yet differentiate to form and core, mantle, and crust.

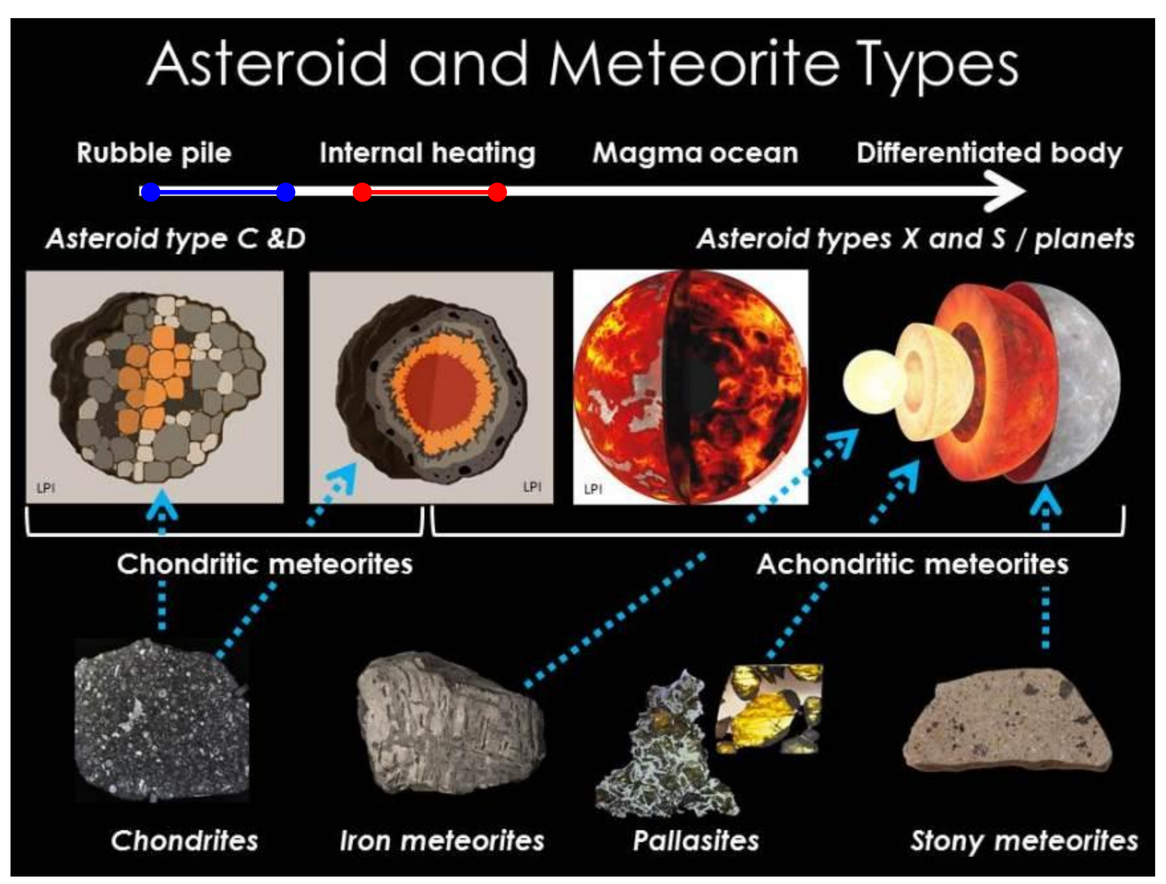

Figure 1. Asteroid and meteorite types relative to planetesimal formation. The red line indicates a suggested lithological range that the NWA ? sample may lie within. The blue line indicates a suggested lithological range that the NWA 5717 type III reference chondrite may lie within. *Image by WordPress.*

Background

- Chondrites are a type of stony meteorite, representing the most primitive material in the solar system.
  - Composed primarily of silicate minerals like olivine, pyroxene, and plagioclase, along with minor amounts of metallic iron and nickel.
- Chondrites are distinguished by the presence of chondrules, small spheroidal objects that formed as molten or partially molten droplets in the early solar nebula.
- Classification of chondrites is based on their petrologic type, a 3-6 scale, which reflects their degree of thermal metamorphism
  - Three main groups of ordinary chondrites:
    - *H:* (high metal content, mostly iron in composition)
    - *L:* (low metal content, but high iron; most common)
    - *LL:* (both low metal and low iron content)
- Chondrites are remnants from the early solar system, providing direct evidence of the processes that occurred during their formation (Fig. 1).
- The meteorite that is the subject of this study was purchased on eBay from a meteorite dealer based in China for use as a classroom teaching sample. It was identified as "NWA" (northwest Africa) but with no other identifying information.
- We are applying basic principles of petrographic and meteoritic analysis to learn more about the significance of this particular meteorite specimen.

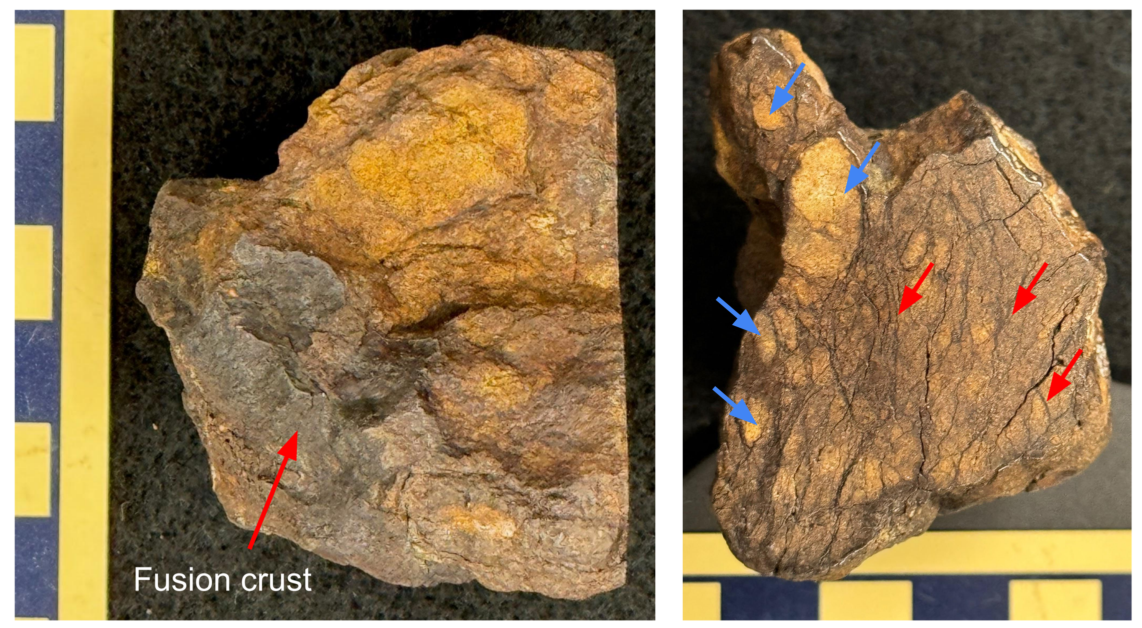

Figure 2. NWA ? meteorite with cut and polished face. The surface is orange to brown (oxidized) with areas of black fusion crust from atmospheric heating. Internal features include rounded fragments with sharp to indistinct rims (blue arrows) and mineral-filled fractures (red arrows.) Scale in centimeters.

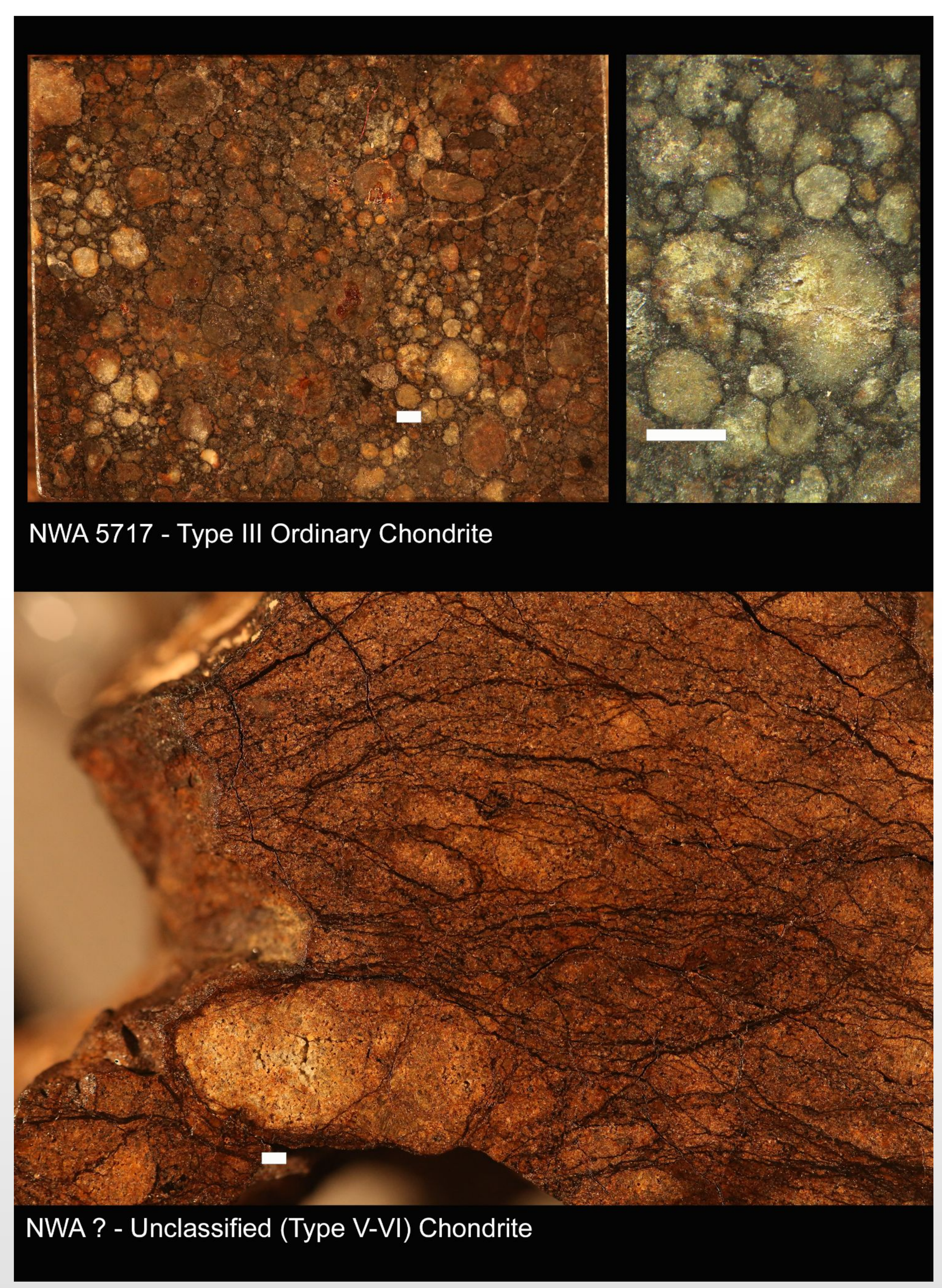

Figure 3. Comparison between NWA 5717 - a Type III ordinary chondrite and NWA ? mystery meteorite. Note the distinct chondrules in NWA 5717 with dark, sulfide rims. In contrast, NWA ? lacks distinct chondrules and is composed of larger oval fragments that vary from having indistinct boundaries to having distinct sulfide rims. Note also the highly fractured texture of NWA ? with fractures infilled by mineral. Scale bars = 1 mm

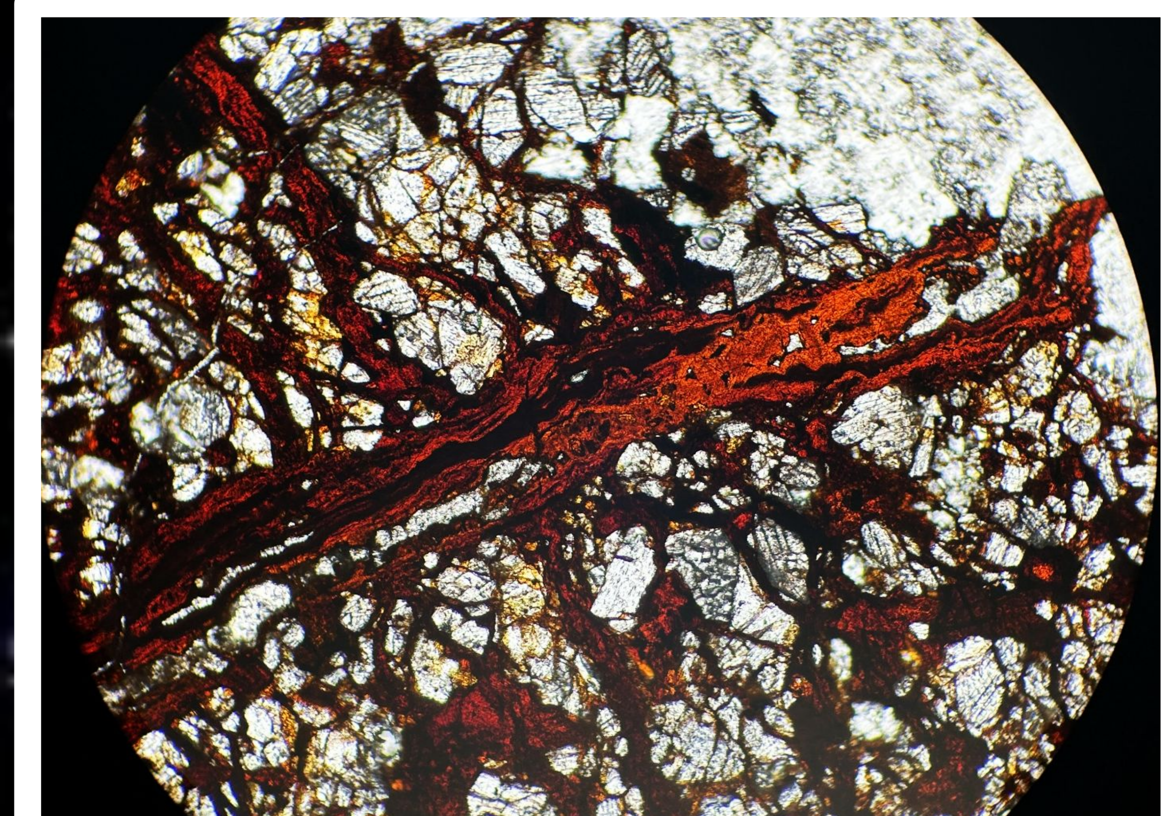

Figure 4. Petrographic thin section image (40x magnification) showing fractures infilled with red material, likely iron oxides, running through the sample. A degree of oxide staining on some crystal grains adjacent to bands is visible. Contrast increased by 30% to enhance detail.

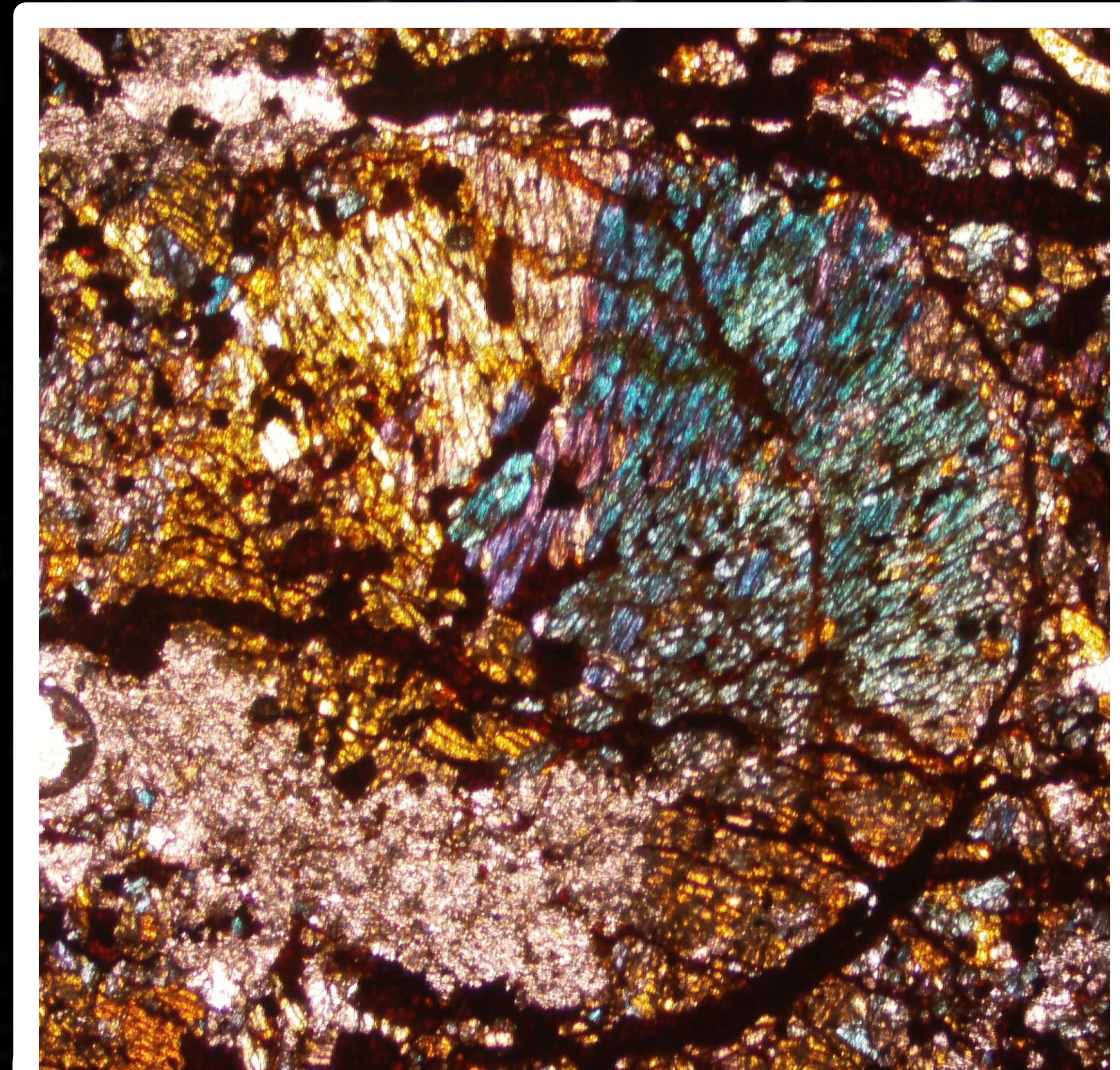

Figure 5. Petrographic thin section image (40x magnification) showing a relic pyroxene chondrule observed in the NWA ? specimen. Several oxide cracks run through the chondrule where it has shattered, and a large white void (now occupied by epoxy) is visible in the bottom left of the chondrule. The original boundary of the chondrule is revealed by the remnant sulfide rim and the oriented pyroxene crystals seen in a radiating pattern.

Methodology

1. The bulk meteorite was cut and polished, creating a fragment for making petrographic thin sections.
2. The fragment was embedded in deep-pour epoxy to provide structural integrity during preparation.
3. The sample was cut to the appropriate size for thin-section preparation using a standard rock saw.
4. Several thin sections were prepared by sanding the face of the sample with progressively finer grit, then adhering the face of the sample to a glass slide.
5. The adhered sample was polished down to transparency for analysis using a petrographic microscope.
6. Completed thin sections were observed under polarized and non-polarized light to assess mineral content and structure.

Results

- The surface of the NSW ? meteorite (Fig .2) is texturally consistent with fusion crust created by melting while falling through the atmosphere (Scott, 2007).
- The measured density of the whole sample is 3.04 g/cm^3, which is statistically low for ordinary chondrites (Moore et. al., 1978).
- The texture of the sample consists of rounded masses of mineral material that range from distinct to indistinct. The entire meteorite is highly fractured with fractures infilled by iron oxide (Fig. 2-4).
- The petrography of the sample is comprised of clusters of angular mineral grains, mostly pyroxene and olivine as indicated by their optical properties.
  - As seen in Figure 4 angular mineral grains dominate the composition.
- Very few intact chondrules were identified; a majority of the structure of the meteorite has been altered significantly by thermal metamorphism, with apparent cracks between mineral grains being mostly filled with iron oxides (Fig. 2-4).
  - One distinct relic chondrule was observed in thin section, identified by a distinct sulfide boundary and radial pattern of pyroxene crystals. This chondrule is extremely weathered, with oxide veins running throughout (Fig 5). This is an indication of terrestrial weathering (Scott, 2007).

Conclusions

- The specimen is provisionally categorized as an ungrouped, equilibrated (metamorphosed) ordinary chondrite, type V-VI.
  - Elemental analysis is required to group the meteorite within the H-L-LL system.
  - Because there are little to no distinguishable chondrules, the material is considered equilibrated..
  - Equilibration is the result of significant metamorphism, placing this specimen on the V-VI range of the type spectrum.
- Absence of intact chondrules is indicative of metamorphism (Scott, 2007).
- Substantial weathering and disintegration of mineral grains.
  - The presence of cracks (Fig 2) and iron oxide bands (Fig 4) throughout the sample is indicative of significant terrestrial weathering. Given that this meteorite's fall was not observed, it may potentially have been exposed to the elements for a very long time.
- A chondrite specimen with well-preserved chondrules is shown in Figure 3. By comparing this sample to the study specimen, the degree of thermal alteration experienced by the latter is apparent.
  - The material of the specimen has experienced compression, and the cracks in the specimen run approximately perpendicular to that force.
- Future work:
  - Analysis by SEM-EDS to fully detail chemical composition and group.

References

- Moore, C.B., Lowenstein, M.Z., and Sipiera, P.P., 1978, Specific Gravity Measurements for Ordinary Chondrites: Meteoritics, v. 13, p. 568.
- Scott, E.R.D., 2007, Chondrites and the Protoplanetary Disk: Annual Review of Earth and Planetary Sciences, v. 35, p. 577–620, doi: 10.1146/annurev.earth.35.031306.140100.

Acknowledgements

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Source (Google Slides):

https://docs.google.com/presentation/d/1c5d3LwVZCgGbSlbZB2c0EU0GvP1HrHX_2xZj4SPB−Iw/edit

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